

| 1.       | Define burst error. Briefly explain a two-dimensional parity check code for error detection.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5        |          |          |          |          |          |          |          |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|------|---------|---|
| 2.       | <p>Suppose, you are provided the following Hamming code C(7,4) for error correction.</p> <table><tr><th>Dataword</th><th>Codeword</th><th>Dataword</th><th>Codeword</th><th>Dataword</th><th>Codeword</th><th>Dataword</th><th>Codeword</th></tr><tr><td>0000</td><td>0000000</td><td>0100</td><td>0100011</td><td>1000</td><td>1000110</td><td>1100</td><td>1100101</td></tr><tr><td>0001</td><td>0001101</td><td>0101</td><td>0101110</td><td>1001</td><td>1001011</td><td>1101</td><td>1101000</td></tr><tr><td>0010</td><td>0010111</td><td>0110</td><td>0110100</td><td>1010</td><td>1010001</td><td>1110</td><td>1110010</td></tr><tr><td>0011</td><td>0011010</td><td>0111</td><td>0111001</td><td>1011</td><td>1011100</td><td>1111</td><td>1111111</td></tr></table> <p>Explain the following cases using Hamming coding at the receiver side:</p> <ol style="list-style-type: none"><li>The codeword 0100011 is received.</li><li>The codeword 0011001 is received.</li><li>The codeword 0001000 is received.</li></ol> | Dataword | Codeword | Dataword | Codeword | Dataword | Codeword | Dataword | Codeword | 0000 | 0000000 | 0100 | 0100011 | 1000 | 1000110 | 1100 | 1100101 | 0001 | 0001101 | 0101 | 0101110 | 1001 | 1001011 | 1101 | 1101000 | 0010 | 0010111 | 0110 | 0110100 | 1010 | 1010001 | 1110 | 1110010 | 0011 | 0011010 | 0111 | 0111001 | 1011 | 1011100 | 1111 | 1111111 | 9 |
| Dataword | Codeword                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Dataword | Codeword | Dataword | Codeword | Dataword | Codeword |          |          |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |   |
| 0000     | 0000000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0100     | 0100011  | 1000     | 1000110  | 1100     | 1100101  |          |          |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |   |
| 0001     | 0001101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0101     | 0101110  | 1001     | 1001011  | 1101     | 1101000  |          |          |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |   |
| 0010     | 0010111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0110     | 0110100  | 1010     | 1010001  | 1110     | 1110010  |          |          |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |   |
| 0011     | 0011010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0111     | 0111001  | 1011     | 1011100  | 1111     | 1111111  |          |          |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |   |
| 3.       | Suppose CRC coding is used in a data communication system where the sender and the receiver agreed to use 1011 as the divisor. If the sender sends 1001 and the receiver receives 1000110, apply the CRC coding system for error detection in this case.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6        |          |          |          |          |          |          |          |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |   |

**Q1: Define burst error. Briefly explain a two-dimensional parity check code for error detection.**

#### **Burst Error:**

A burst error occurs when two or more bits in a data sequence are altered due to noise or interference. Unlike a single-bit error, a burst error affects multiple bits in a sequence, typically in a contiguous manner.

#### **Two-Dimensional Parity Check:**

A two-dimensional parity check is an error detection method that organizes data into a grid format and applies parity checks to both rows and columns.

#### **Steps:**

- 1. Arrange Data in a Grid:** Place data bits in a 2D matrix.
- 2. Calculate Row Parity:** Compute parity for each row and add it as an extra bit.
- 3. Calculate Column Parity:** Compute parity for each column and add it as an extra bit.
- 4. Send the Matrix:** Transmit the entire matrix, including row and column parities.
- 5. Error Detection:** At the receiver end, recompute parity and compare with received parities to identify errors.

## Solution for Hamming Code Error Detection and Correction

### Understanding the Positions in Hamming (7,4) Code

Hamming (7,4) code consists of 7-bit codewords with:

- 3 Parity Bits:  $P_0, P_1, P_2$
- 4 Data Bits:  $B_0, B_1, B_2, B_3$

**Bit Positions in Codeword:**

| Position | 1     | 2     | 3     | 4     | 5     | 6     | 7     |
|----------|-------|-------|-------|-------|-------|-------|-------|
| Bit Type | $P_0$ | $P_1$ | $B_2$ | $P_2$ | $B_0$ | $B_3$ | $B_1$ |

### Parity Bits Cover These Positions:

- $P_0$  covers  $\rightarrow 1, 3, 5, 7 \rightarrow (P_0, B_2, B_0, B_1)$
- $P_1$  covers  $\rightarrow 2, 3, 6, 7 \rightarrow (P_1, B_2, B_3, B_1)$
- $P_2$  covers  $\rightarrow 4, 5, 6, 7 \rightarrow (P_2, B_0, B_3, B_1)$

### (i) Received Codeword: 0001011

#### Step 1: Identify the Parity Bits

From 0001011, the bits are assigned as follows:

| Position | 1     | 2     | 3     | 4     | 5     | 6     | 7     |
|----------|-------|-------|-------|-------|-------|-------|-------|
| Bit Type | $P_0$ | $P_1$ | $B_2$ | $P_2$ | $B_0$ | $B_3$ | $B_1$ |
| Received | 0     | 0     | 0     | 1     | 0     | 1     | 1     |

#### Step 2: Compute Parity Check Syndrome

Using the parity bit rules:

- $S_0 = P_0 \oplus B_2 \oplus B_0 \oplus B_1$   
 $\rightarrow S_0 = 0 \oplus 0 \oplus 0 \oplus 1 = 1$
- $S_1 = P_1 \oplus B_2 \oplus B_3 \oplus B_1$   
 $\rightarrow S_1 = 0 \oplus 0 \oplus 1 \oplus 1 = 0$
- $S_2 = P_2 \oplus B_0 \oplus B_3 \oplus B_1$   
 $\rightarrow S_2 = 1 \oplus 0 \oplus 1 \oplus 1 = 1$

#### Step 3: Determine the Error Position

The syndrome is  $S_2 S_1 S_0 = 101$  (decimal 5).

So, the error is at position 5 ( $B_0$ ).

#### Step 4: Correct the Error

Flip  $B_0$  ( $0 \rightarrow 1$ ), so the corrected codeword is:

0001111

**(ii) Received Codeword: 0011001**

**Step 1: Identify the Parity Bits**

From 0011001, the bits are assigned as follows:

| Position | 1     | 2     | 3     | 4     | 5     | 6     | 7     |
|----------|-------|-------|-------|-------|-------|-------|-------|
| Bit Type | $P_0$ | $P_1$ | $B_2$ | $P_2$ | $B_0$ | $B_3$ | $B_1$ |
| Received | 0     | 0     | 1     | 1     | 0     | 0     | 1     |

**Step 2: Compute Parity Check Syndrome**

Using the parity bit rules:

- $S_0 = P_0 \oplus B_2 \oplus B_0 \oplus B_1$   
 $\rightarrow S_0 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$
- $S_1 = P_1 \oplus B_2 \oplus B_3 \oplus B_1$   
 $\rightarrow S_1 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$
- $S_2 = P_2 \oplus B_0 \oplus B_3 \oplus B_1$   
 $\rightarrow S_2 = 1 \oplus 0 \oplus 0 \oplus 1 = 0$

**Step 3: Determine the Error Position**

The syndrome is  $S_2 S_1 S_0 = 000$ .

Since all bits are 0, there is no error.

So, the received codeword 0011001 is already correct.

c)

